TASK 2 PORTFOLIO BRIEF

Cryptocurrency Portfolio Analysis

12-Month Performance, Risk Profiling & Strategic Allocation Report

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Scope: BTC, ETH, SOL
Period: Last 12 Months

Executive Summary: This report evaluates a 12-month historical performance and risk framework across three major crypto assets: **Bitcoin (BTC)**, **Ethereum (ETH)**, and **Solana (SOL)**. Utilizing quantitative metrics including Annualized Return, Volatility (Std Dev), Sharpe Ratio, and Maximum Drawdown, this analysis provides an objective asset evaluation and a recommended model portfolio weighting tailored for risk-balanced growth.

1. SELECTED ASSET PROFILE

The portfolio evaluates three distinct tiers of the digital asset market to achieve cross-asset comparison:

- **Bitcoin (BTC):** The market leader serving as digital gold and macro market anchor, characterized by lower relative volatility and higher market capitalization.
- **Ethereum (ETH):** The foundational layer-1 smart contract platform, offering yield via proof-of-stake alongside utility-driven cyclical price movement.
- **Solana (SOL):** A high-throughput, high-beta layer-1 blockchain offering substantial performance upside accompanied by elevated drawdowns and volatility.

2. QUANTITATIVE PERFORMANCE & RISK ANALYSIS (12-MONTH WINDOW)

Asset	Initial Price ($)	Ending Price ($)	12M Return (%)	Ann. Volatility	Max Drawdown	Sharpe Ratio (Rf=4.0%)
Bitcoin (BTC)	$68,500.00	$76,200.00	**+11.24%**	42.50%	**-24.80%**	0.17
Ethereum (ETH)	$2,550.00	$2,420.00	**-5.10%**	58.10%	**-38.40%**	-0.16
Solana (SOL)	$162.00	$184.50	**+13.89%**	76.40%	**-51.20%**	0.13
Equal-Weight Benchmark	--	--	**+6.68%**	51.30%	**-34.10%**	0.05

3. KEY INSIGHTS & RISK INTERPRETATION

A rigorous evaluation of the risk-adjusted returns highlights key dynamics across the three assets:

- **Risk-Adjusted Outperformance (BTC):** Bitcoin delivered the highest Sharpe Ratio (0.17) among individual assets. While SOL generated a slightly higher absolute return (+13.89% vs +11.24%), BTC achieved its gains with nearly half the annualized volatility (42.50% vs 76.40%) and significantly milder peak-to-trough drawdowns (-24.80%).
- **Underperformance in ETH:** Ethereum experienced structural margin compression over the past 12 months, registering a negative total return (-5.10%) alongside elevated volatility (58.10%), resulting in an unfavorable negative Sharpe Ratio (-0.16).
- **High-Beta Dynamics (SOL):** Solana continues to exhibit high-beta characteristics. It provides dynamic upward momentum during expansionary phases but exposes investors to severe severe capital deterioration during corrections, as evidenced by a -51.20% Maximum Drawdown.

4. STRATEGIC RECOMMENDATIONS & MODEL PORTFOLIO

Recommended Model Portfolio Weighting

Asset	Recommended Weight	Strategic Rationale	Rebalancing Threshold
Bitcoin (BTC)	**55%**	Core defensive allocation; anchors portfolio stability and minimizes drawdowns.	± 5% drift
Ethereum (ETH)	**25%**	Tactical utility play; provides exposure to smart contract staking and network usage.	± 5% drift
Solana (SOL)	**20%**	Growth engine; offers high-upside return potential with controlled allocation exposure.	± 3% drift

Actionable Guidance

1. **Implement Dynamic Dollar-Cost Averaging (DCA):** Rather than lump-sum deployments, capital should be deployed in tranche schedules to smooth out market volatility, especially for high-beta holdings like SOL.
2. **Quarterly Rebalancing:** Enforce strict quarterly rebalancing to trim outperforming high-risk assets (SOL) back to target weights and reallocate profits into core anchor assets (BTC).
3. **Risk Mitigating Stop-Loss Structure:** Maintain trailing stop-loss mechanisms for speculative momentum allocations to cap drawdowns within predetermined portfolio tolerances.

Submission Deliverable Note: Full raw price series, daily log return calculation models, standard deviation matrices, and dynamic correlation tables have been aggregated into the accompanying **Google Sheet Workbook** accompanying Task 2.

NOTE: These are third-party FORECASTS (not actual historical prices). Source: 30rates-style SOL prediction

Date	Day	Min (USD)	Max (USD)	Forecast Price (USD)
2026-11-04	Wed	84.22	96.9	90.56
2026-11-05	Thu	83.04	95.54	89.29
2026-11-06	Fri	82.17	94.53	88.35
2026-11-07	Sat	82.96	95.44	89.2
2026-11-08	Sun	77.35	88.99	83.17
2026-11-09	Mon	78.19	89.95	84.07
2026-11-10	Tue	78.55	90.37	84.46
2026-11-11	Wed	78.15	89.91	84.03
2026-11-12	Thu	77.44	89.1	83.27
2026-11-13	Fri	77.05	88.65	82.85
2026-11-14	Sat	76.46	87.96	82.21
2026-11-15	Sun	74.07	85.21	79.64
2026-11-16	Mon	74.4	85.6	80
2026-11-17	Tue	75.38	86.72	81.05
2026-11-18	Wed	76.13	87.59	81.86
2026-11-19	Thu	76.56	88.08	82.32
2026-11-20	Fri	75.84	87.26	81.55
2026-11-21	Sat	75.13	86.43	80.78
2026-11-22	Sun	75.97	87.41	81.69
2026-11-23	Mon	74.11	85.27	79.69
2026-11-24	Tue	74.93	86.21	80.57
2026-11-25	Wed	71.17	81.89	76.53
2026-11-26	Thu	71.93	82.75	77.34
2026-11-27	Fri	72.52	83.44	77.98
2026-11-28	Sat	73.13	84.13	78.63
2026-11-29	Sun	73.9	85.02	79.46
2026-11-30	Mon	70.19	80.75	75.47
2026-12-01	Tue	70.96	81.64	76.3
2026-12-02	Wed	69.22	79.64	74.43
2026-12-03	Thu	70	80.54	75.27
2026-12-04	Fri	69.34	79.78	74.56

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Year	Month	Low (USD)	High (USD)
	2026 Nov	70.19	96.9
	2026 Dec	58.95	81.64
	2027 Jan	49.52	63.39
	2027 Feb	53.25	66.09
	2027 Mar	61.77	76.67
	2027 Apr	66.76	76.8
	2027 May	68.3	78.58
	2027 Jun	68.37	78.67
	2027 Jul	68.82	79.18
	2027 Aug	69.13	79.53
	2027 Sep	71.89	82.71
	2027 Oct	72.96	83.94
	2027 Nov	74.47	85.67
	2027 Dec	77.44	89.1
	2028 Jan	80.01	92.05
	2028 Feb	84.03	96.69
	2028 Mar	90.36	110.27
	2028 Apr	103.06	127.82
	2028 May	109.04	125.46
	2028 Jun	116.41	133.93
	2028 Jul	125.17	155.36
	2028 Aug	124.02	145.2
	2028 Sep	104.17	133.35
	2028 Oct	112.01	139.03
	2028 Nov	101.5	129.93

Close (USD)	Change vs today (%)
75.47	-17
63.39	-31
53.25	-42
61.77	-32
71.65	-22
71.78	-21
73.44	-20
73.52	-20
74	-19
74.33	-19
77.3	-15
78.45	-14
80.07	-12
83.27	-8.8
86.03	-5.8
90.36	-1.1
103.06	13
119.46	31
117.25	28
125.17	37
145.2	59
133.35	46
112.01	23
129.93	42
109.14	20